RDS203_FinalProject

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```r
lib <- c("dplyr", "tidyr", "ggplot2", "gghighlight", "ggrepel", "gt")

for(pkg in lib){
  if(!require(pkg, character.only = TRUE)){
    install.packages(pkg)
    library(pkg, character.only = TRUE)
  }
}
```

```
## Loading required package: dplyr

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

## Loading required package: tidyr

## Loading required package: ggplot2

## Loading required package: gghighlight

## Loading required package: ggrepel

## Loading required package: gt
```

1. Result Overview

```r
res <- read.csv("./FinalProject_Code/output/results.csv")
head(res, 3)
```

```
##    strategy startAge stopAge frequency cost    qaly lifeYears crcDeath
## 1 S40_69_F1       40      69         1  900 44.6580        45        0
## 2 S40_69_F1       40      69         1  900 40.1196        45        0
## 3 S40_69_F1       40      69         1  830 21.6580        22        0
##   incidentCRC mortalityReduction
## 1           0          0.1788991
## 2           1          0.1788991
## 3           0          0.1788991
```

1.1 Average performance

```r
# Compute average
avg_df <- res|>
  group_by(strategy)|>
  summarise(cost_avg = mean(cost),
            qaly_avg = mean(qaly),
            mortalityReduction_avg = mean(mortalityReduction))|>
  mutate(start_age = as.numeric(sub("S(\\d+)_(\\d+)_F(\\d+)", "\\1", strategy)),
         stop_age = as.numeric(sub("S(\\d+)_(\\d+)_F(\\d+)", "\\2", strategy)),
         frequency = as.numeric(sub("S(\\d+)_(\\d+)_F(\\d+)", "\\3", strategy)))|>
  mutate(frequency = recode(frequency,
                            "1" = "Annual",
                            "2" = "Biennial"))

avg_cols <- colnames(avg_df)[grep("_avg$", colnames(avg_df))]
avg_df <- avg_df|>
  mutate(across(all_of(avg_cols), ~ round(.x, digits = 2)))

avg_df
```

```
## # A tibble: 18 x 7
##    strategy  cost_avg qaly_avg mortalityReduction_avg start_age stop_age
##    <chr>        <dbl>    <dbl>                  <dbl>     <dbl>    <dbl>
##  1 S40_69_F1    1535.     37.7                   0.18        40       69
##  2 S40_69_F2     986.     37.9                   0.23        40       69
##  3 S40_74_F1    1872.     37.6                   0.33        40       74
##  4 S40_74_F2    1272.     37.9                   0.29        40       74
##  5 S40_84_F1    2506.     37.6                   0.33        40       84
##  6 S40_84_F2    1766.     37.8                   0.44        40       84
##  7 S45_69_F1    1359.     37.8                   0.21        45       69
##  8 S45_69_F2     945.     37.9                   0.27        45       69
##  9 S45_74_F1    1677.     37.7                   0.25        45       74
## 10 S45_74_F2    1134.     37.9                   0.35        45       74
## 11 S45_84_F1    2321.     37.6                   0.4         45       84
## 12 S45_84_F2    1643.     37.8                   0.39        45       84
## 13 S50_69_F1    1202.     37.8                   0.25        50       69
## 14 S50_69_F2     809.     38.0                   0.26        50       69
## 15 S50_74_F1    1513.     37.8                   0.34        50       74
## 16 S50_74_F2    1097.     38.0                   0.31        50       74
## 17 S50_84_F1    2153.     37.7                   0.3         50       84
## 18 S50_84_F2    1591.     37.9                   0.33        50       84
## # i 1 more variable: frequency <chr>
```

1.2 Best performance

```r
# Find best value
cost_best <- avg_df|>
  slice_min(cost_avg)|>
  mutate(category = "Lowest costs")

qaly_best <- avg_df|>
  slice_max(qaly_avg)|>
  mutate(category = "Highest QALYs")
```

```r
mortalityReduction_best <- avg_df|>
  slice_max(mortalityReduction_avg)|>
  mutate(category = "Greatest CRC mortality reduction")

best_df <- bind_rows(
  cost_best,
  qaly_best,
  mortalityReduction_best
)
best_df
```

```
## # A tibble: 3 x 8
##   strategy cost_avg qaly_avg mortalityReduction_avg start_age stop_age frequency
##   <chr>       <dbl>    <dbl>                  <dbl>     <dbl>    <dbl> <chr>
## 1 S50_69_~     809.     38.0                   0.26        50       69 Biennial
## 2 S50_69_~     809.     38.0                   0.26        50       69 Biennial
## 3 S40_84_~    1766.     37.8                   0.44        40       84 Biennial
## # i 1 more variable: category <chr>
```

```r
best_strategy <- best_df$strategy
compare_df <- res|>
  filter(strategy %in% best_strategy)

t.test(cost ~ strategy, data = compare_df)
```

```
##
##  Welch Two Sample t-test
##
## data:  cost by strategy
## t = 64.214, df = 175839, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group S40_84_F2 and group S50_69_F2 is not
## 95 percent confidence interval:
##  927.1302 985.5086
## sample estimates:
## mean in group S40_84_F2 mean in group S50_69_F2
##                1765.667                 809.348
```

```r
t.test(qaly ~ strategy, data = compare_df)
```

```
##
##  Welch Two Sample t-test
##
## data:  qaly by strategy
## t = -3.5705, df = 199979, p-value = 0.0003563
## alternative hypothesis: true difference in means between group S40_84_F2 and group S50_69_F2 is not
## 95 percent confidence interval:
##  -0.24165058 -0.07037233
## sample estimates:
## mean in group S40_84_F2 mean in group S50_69_F2
##                37.80317                37.95918
```

```r
best_tbl <- best_df|>
  select(start_age, stop_age, frequency, category, avg_cols)|>
  arrange(category)|>
```

Start Age (years)	Stop Age (years)	Frequency	Performance Metric	Average Cost (USD)
40	84	Biennial	Greatest CRC mortality reduction	1765.67
50	69	Biennial	Highest QALYs	809.35
50	69	Biennial	Lowest costs	809.35

```r
  gt()|>
  cols_label(
    start_age = "Start Age (years)",
    stop_age = "Stop Age (years)",
    frequency = "Frequency",
    category = "Performance Metric",

    cost_avg = "Average Cost (USD)",
    qaly_avg = "Average QALYs",
    mortalityReduction_avg = "Average CRC Mortality Reduction (%)"
  )|>
  opt_table_font(font = "Times New Roman")|>
  tab_style(
    style = cell_text(
      weight = "bold",
      size = px(14),
      align = "center"
      ),
    locations = cells_column_labels()
  )|>
  tab_style(
    style = cell_text(
      align = "center",
      size = px(14)
    ),
    locations = cells_body()
  )
```

```
## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use `all_of()` or `any_of()` instead.
##   # Was:
##   data %>% select(avg_cols)
##
##   # Now:
##   data %>% select(all_of(avg_cols))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```r
best_tbl
```

```r
gtsave(data = best_tbl, filename = "best_tbl.png", path = "./table")
```

```
## file:////var/folders/10/f5shtmxx0jl9dg525f1vctqm0000gn/T//RtmpLwnkV1/file490614b638e4.html screensho
```

```r
avg_tbl <- avg_df|>
  select(strategy, start_age, stop_age, frequency, avg_cols)|>
  gt()|>

  tab_style(
    style = cell_fill(color = "lightblue"),
    locations = cells_body(
      rows = strategy %in% best_df$strategy)
  )|>
  cols_hide(strategy)|>
  tab_source_note(
    source_note = md(
      "Highlighted rows represent best-performing screening strategies across selected outcome measures
    )
  )|>

  cols_label(
    start_age = "Start Age (years)",
    stop_age = "Stop Age (years)",
    frequency = "Frequency",

    cost_avg = "Average Cost (USD)",
    qaly_avg = "Average QALYs",
    mortalityReduction_avg = "Average Mortality Reduction (%)"
  )|>

  opt_table_font(font = "Times New Roman")|>
  tab_style(
    style = cell_text(
      weight = "bold",
      size = px(14),
      align = "center"
      ),
    locations = cells_column_labels()
  )|>
  tab_style(
    style = cell_text(
      align = "center",
      size = px(14),
    ),
    locations = cells_body()
  )

avg_tbl
```

```r
gtsave(data = avg_tbl, filename = "avg_tbl.png", path = "./table")
```

```
## file:////var/folders/10/f5shtmxx0jl9dg525f1vctqm0000gn/T//RtmpLwnkV1/file49067ecff410.html screensho
```

1.2 Worst performance

```r
# Find best value
cost_worst <- avg_df|>
  slice_max(cost_avg)|>
```

Start Age (years)	Stop Age (years)	Frequency	Average Cost (USD)	Average QALYs	Average Mort
40	69	Annual	1534.57	37.68	
40	69	Biennial	985.95	37.87	
40	74	Annual	1872.13	37.60	
40	74	Biennial	1272.10	37.86	
40	84	Annual	2505.97	37.58	
40	84	Biennial	1765.67	37.80	
45	69	Annual	1359.46	37.76	
45	69	Biennial	945.24	37.92	
45	74	Annual	1676.63	37.73	
45	74	Biennial	1134.20	37.88	
45	84	Annual	2320.79	37.65	
45	84	Biennial	1642.96	37.84	
50	69	Annual	1202.08	37.82	
50	69	Biennial	809.35	37.96	
50	74	Annual	1512.91	37.81	
50	74	Biennial	1096.60	37.95	
50	84	Annual	2153.48	37.70	
50	84	Biennial	1590.90	37.88	

Highlighted rows represent best-performing screening strategies across selected outcome measures.

```
  mutate(category = "Highest costs")

qaly_worst <- avg_df|>
  slice_min(qaly_avg)|>
  mutate(category = "Lowest QALYs")

mortalityReduction_worst <- avg_df|>
  slice_min(mortalityReduction_avg)|>
  mutate(category = "Worst mortality reduction")

worst_df <- bind_rows(
  cost_worst,
  qaly_worst,
  mortalityReduction_worst
)
worst_df
```

```
## # A tibble: 3 x 8
##   strategy cost_avg qaly_avg mortalityReduction_avg start_age stop_age frequency
##   <chr>       <dbl>    <dbl>                  <dbl>     <dbl>    <dbl> <chr>
## 1 S40_84_~    2506.     37.6                   0.33        40       84 Annual
## 2 S40_84_~    2506.     37.6                   0.33        40       84 Annual
## 3 S40_69_~    1535.     37.7                   0.18        40       69 Annual
## # i 1 more variable: category <chr>
```

Start Age (years)	Stop Age (years)	Frequency	Performance Metric	Average Cost (USD)	Averag
40	84	Annual	Highest costs	2505.97	3
40	84	Annual	Lowest QALYs	2505.97	3
40	69	Annual	Worst mortality reduction	1534.57	3

```
worst_tbl <- worst_df|>
  select(start_age, stop_age, frequency, category, avg_cols)|>
  arrange(start_age)|>
  gt()|>
  cols_label(
    start_age = "Start Age (years)",
    stop_age = "Stop Age (years)",
    frequency = "Frequency",

    category = "Performance Metric",
    cost_avg = "Average Cost (USD)",
    qaly_avg = "Average QALYs",
    mortalityReduction_avg = "Average Mortality Reduction (%)"
  )|>
  opt_table_font(font = "Times New Roman")|>
  tab_style(
    style = cell_text(
      weight = "bold",
      size = px(14),
      align = "center"
      ),
    locations = cells_column_labels()
  )|>
  tab_style(
    style = cell_text(
      align = "center",
      size = px(14)
    ),
    locations = cells_body()
  )


worst_tbl
```

```
gtsave(data = worst_tbl, filename = "worst_tbl.png", path = "./table")
```

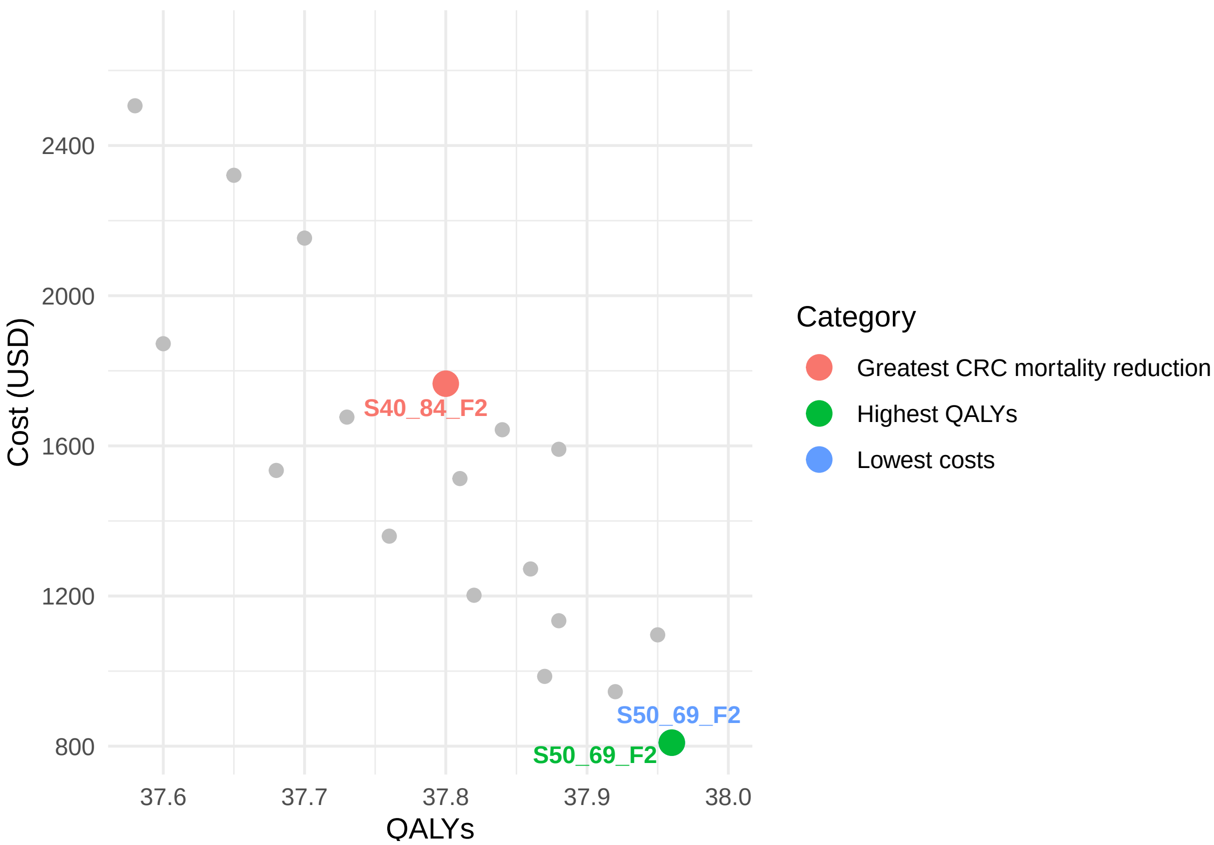
```
## file:////var/folders/10/f5shtmxx0jl9dg525f1vctqm0000gn/T//RtmpLwnkV1/file490647b11b32.html screensho
```

2. Performance Plot

2.1 Best strategy

```
best_label <- best_df|>
  dplyr::rename("Category" = "category")

ggplot(avg_df, aes(x = qaly_avg, y = cost_avg))+
  # all points
```

```r
geom_point(colour = "grey", size = 2)+

# highlight best points (best_label)
geom_point(
  data = best_label,
  aes(colour = Category),
  size = 4)+
geom_text_repel(
  data = best_label,
  aes(label = strategy, colour = Category),
  size = 3,
  show.legend = FALSE,
  max.overlaps = Inf,
  fontface = "bold",
  box.padding = 0.3,
  point.padding = 0.5,
  segment.color = "grey50")+

scale_x_continuous(expand = expansion(mult = c(0.05, 0.15)))+
scale_y_continuous(expand = expansion(mult = c(0.05, 0.15)))+
theme_minimal()+
labs(
  x = "QALYs",
  y = "Cost (USD)"
)
```

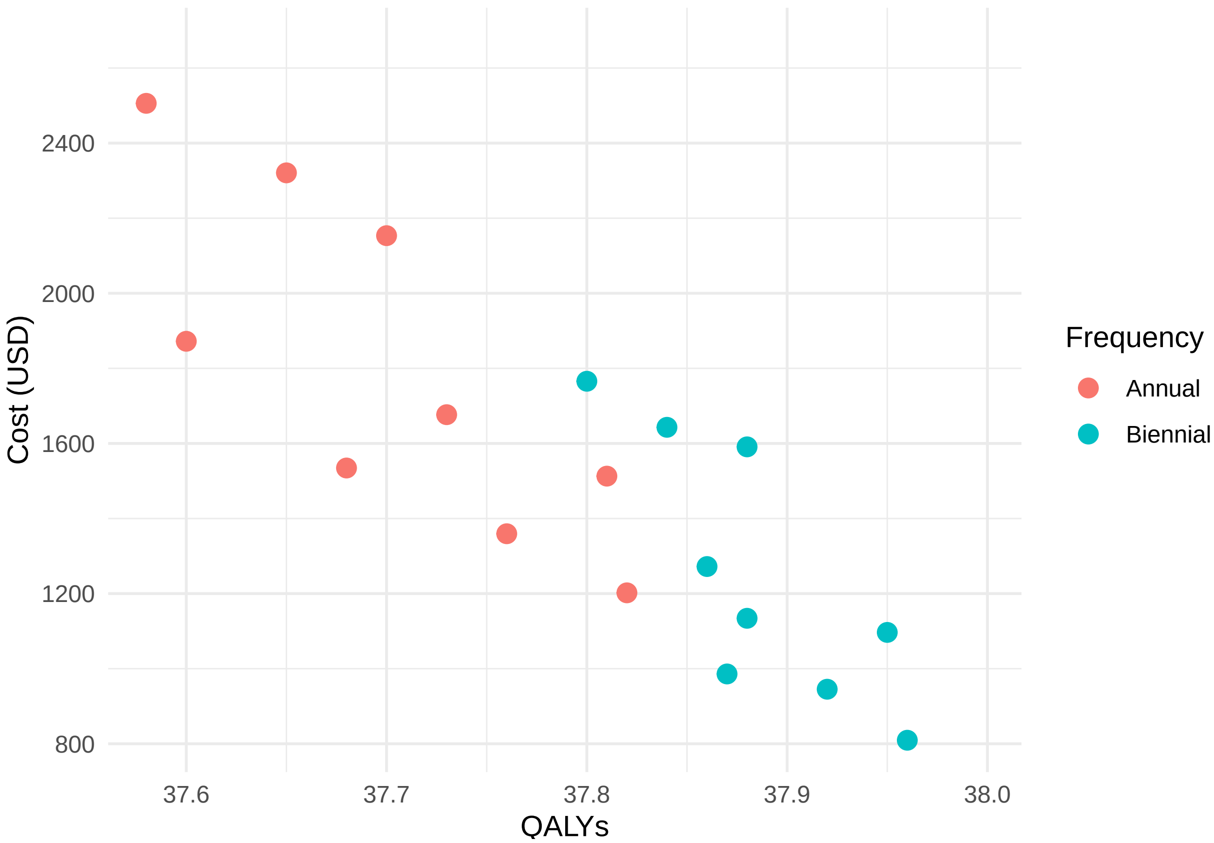


```r
ggsave("./pic/bestStrategy.png")
```

```
## Saving 6.5 x 4.5 in image
```

2.2 Cost vs QALYs by frequency

```r
ggplot(avg_df, aes(x = qaly_avg, y = cost_avg, colour = factor(frequency)))+
  geom_point(size = 3)+
  labs(
    x = "QALYs",
    y = "Cost (USD)",
    colour = "Frequency"
  )+
  scale_x_continuous(expand = expansion(mult = c(0.05, 0.15)))+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.15)))+
  theme_minimal()
```



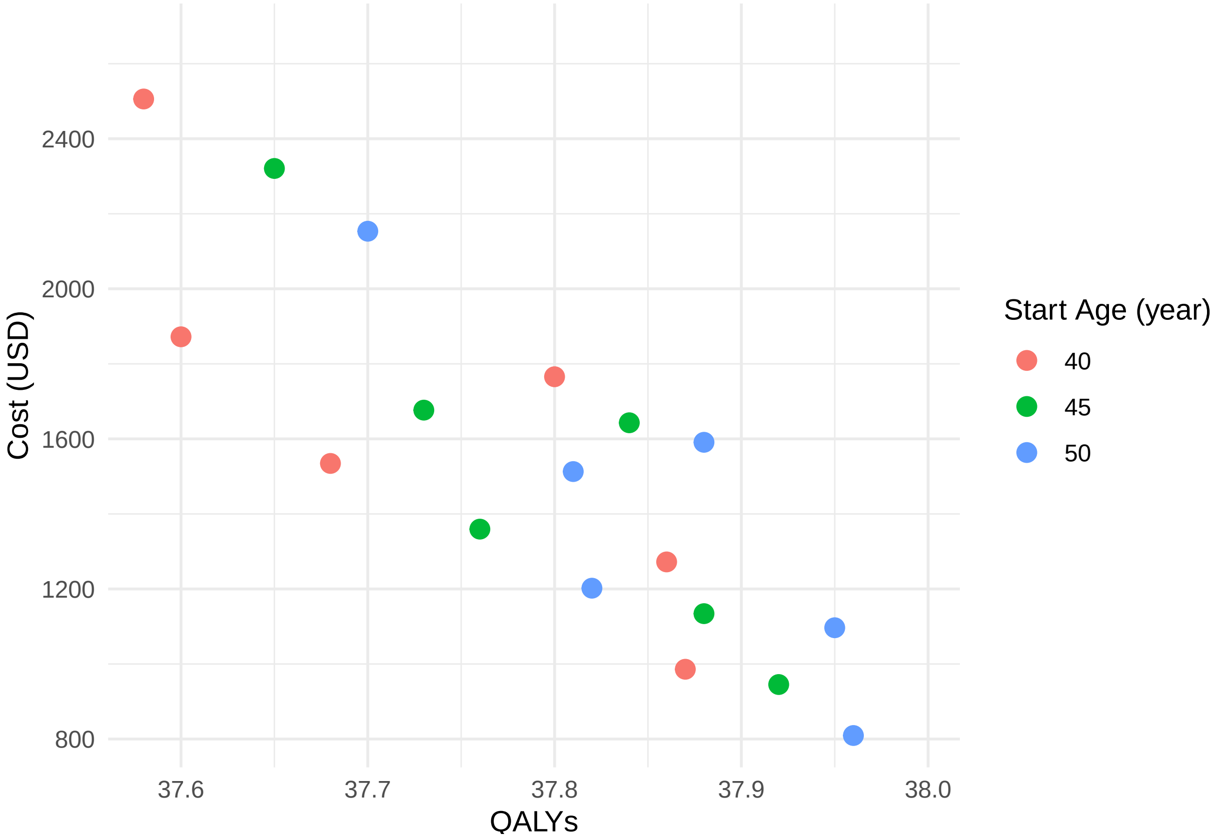

```r
ggsave("./pic/cost_vs_qaly_byFrequency.png")
```

```
## Saving 6.5 x 4.5 in image
```

2.3 Cost vs QALYs by start age

```r
ggplot(avg_df, aes(x = qaly_avg, y = cost_avg, colour = factor(start_age)))+
  geom_point(size = 3)+
  labs(
    x = "QALYs",
```

```
    y = "Cost (USD)",
    colour = "Start Age (year)"
  )+
  scale_x_continuous(expand = expansion(mult = c(0.05, 0.15)))+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.15)))+
  theme_minimal()
```

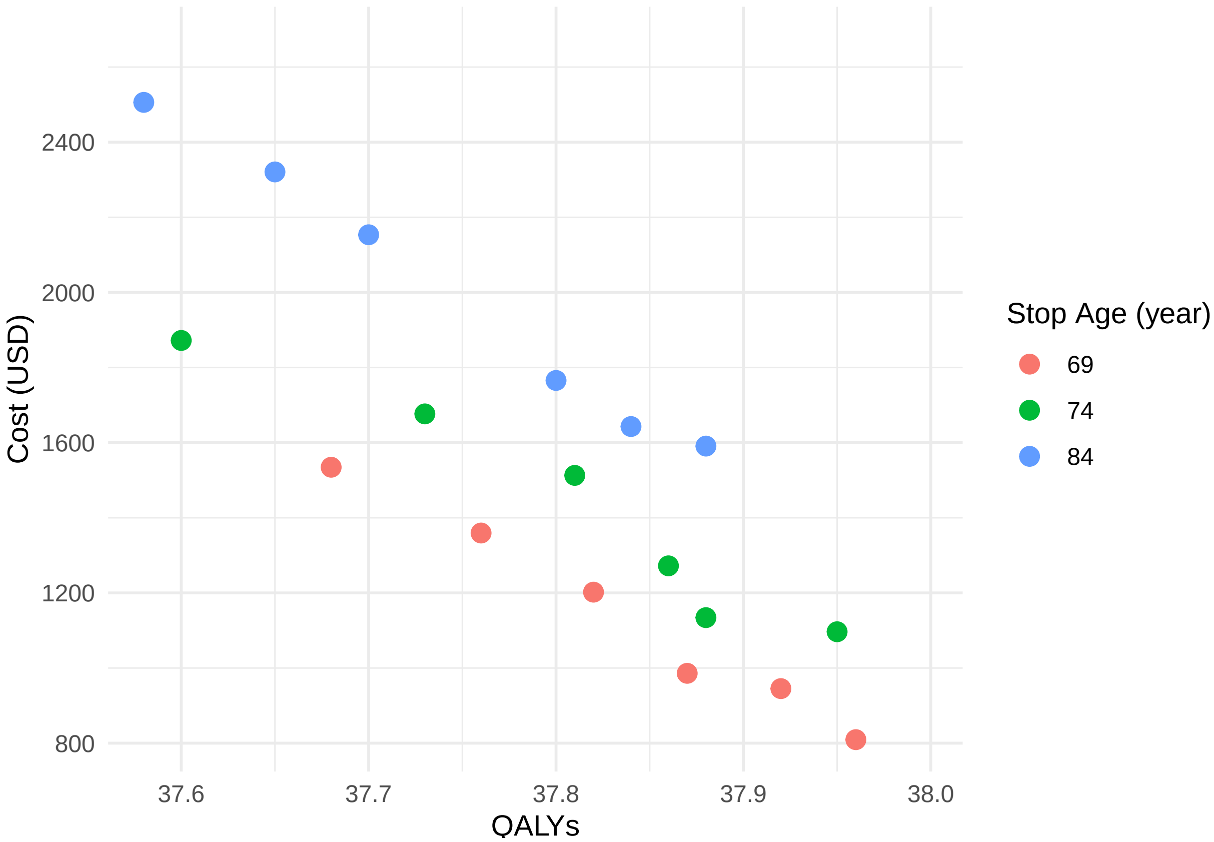



```
ggsave("./pic/cost_vs_qaly_byStartAge.png")
```

```
## Saving 6.5 x 4.5 in image
```

2.4 Cost vs QALYs by stop age

```
ggplot(avg_df, aes(x = qaly_avg, y = cost_avg, colour = factor(stop_age)))+
  geom_point(size = 3)+
  labs(
    x = "QALYs",
    y = "Cost (USD)",
    colour = "Stop Age (year)"
  )+
  scale_x_continuous(expand = expansion(mult = c(0.05, 0.15)))+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.15)))+
  theme_minimal()
```

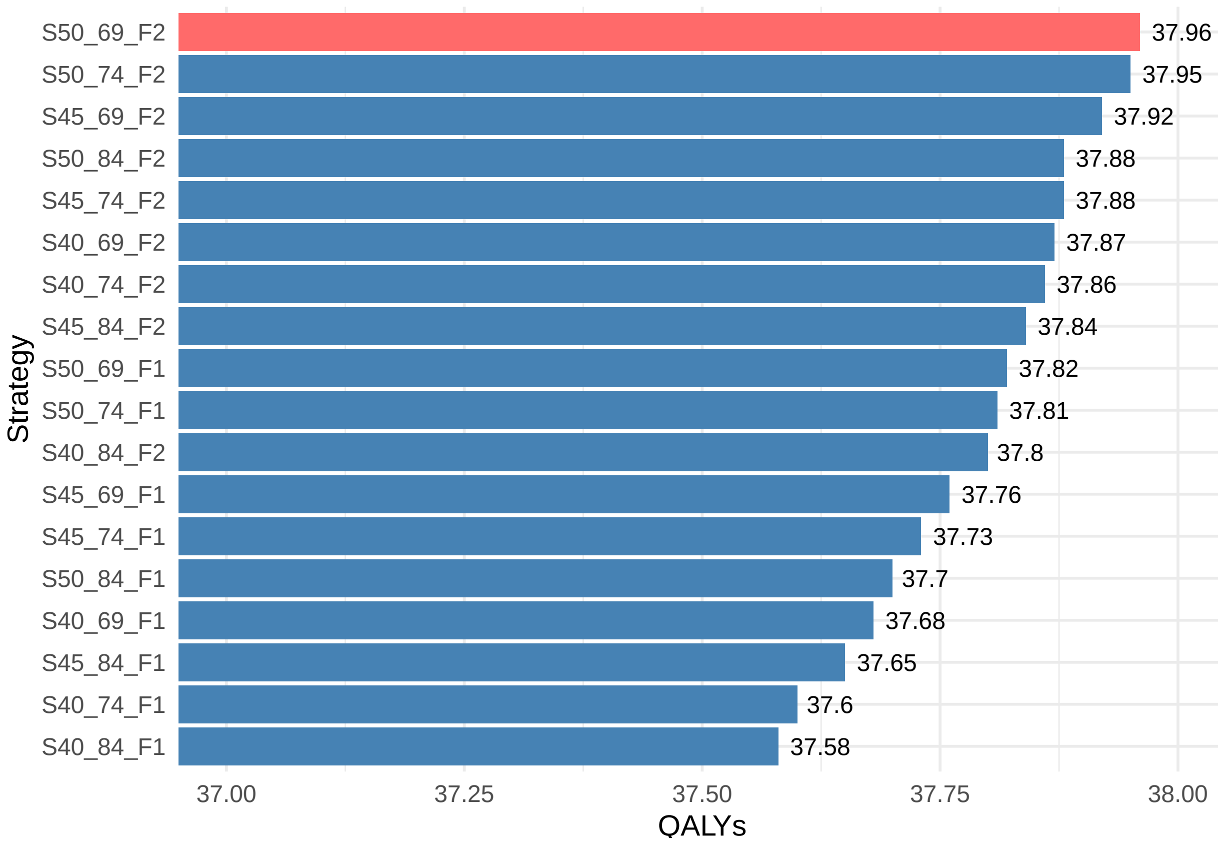



```
ggsave("./pic/cost_vs_qaly_byStopAge.png")
```

```
## Saving 6.5 x 4.5 in image
```

2.5 QALYs by strategy

```
ggplot(avg_df, aes(x = reorder(strategy, qaly_avg), y = qaly_avg))+
  geom_col(aes(fill = qaly_avg == max(qaly_avg)))+
  scale_fill_manual(values = c("steelblue", "#FF6A6A"), guide = "none")+
  coord_flip(ylim = c(37.0, 38.0))+
  geom_text(aes(label = round(qaly_avg, 3)), hjust = -0.2, size = 3)+
  labs(
    x = "Strategy",
    y = "QALYs"
  )+
  theme_minimal()
```

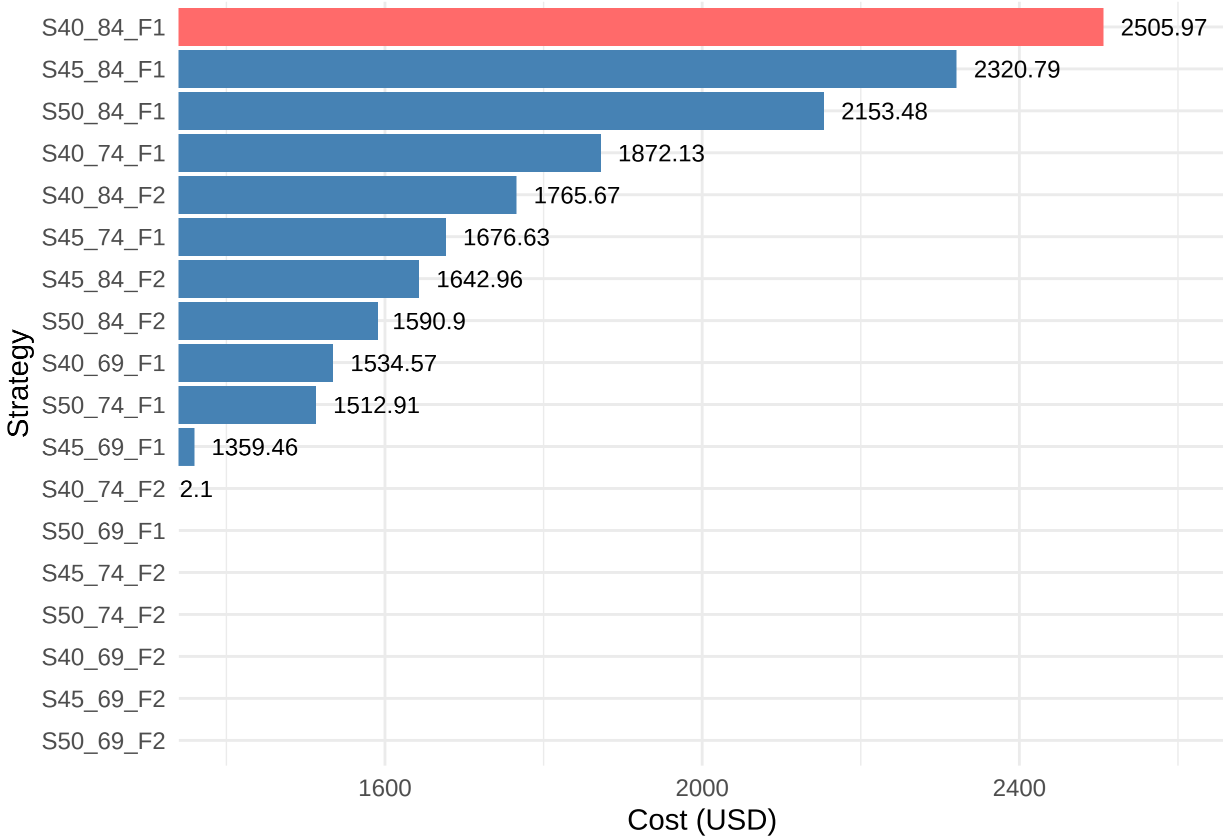



```
ggsave("./pic/qaly_by_strategy.png")
```

```
## Saving 6.5 x 4.5 in image
```

2.6 Cost by strategy

```
ggplot(avg_df, aes(x = reorder(strategy, cost_avg), y = cost_avg))+
  geom_col(aes(fill = cost_avg == max(cost_avg)))+
  scale_fill_manual(values = c("steelblue", "#FF6A6A"),
                    guide = "none")+
  coord_flip(ylim = c(1400, 2600))+
  geom_text(aes(label = round(cost_avg, 3)), hjust = -0.2, size = 3)+
  labs(
    x = "Strategy",
    y = "Cost (USD)"
  )+
  theme_minimal()
```

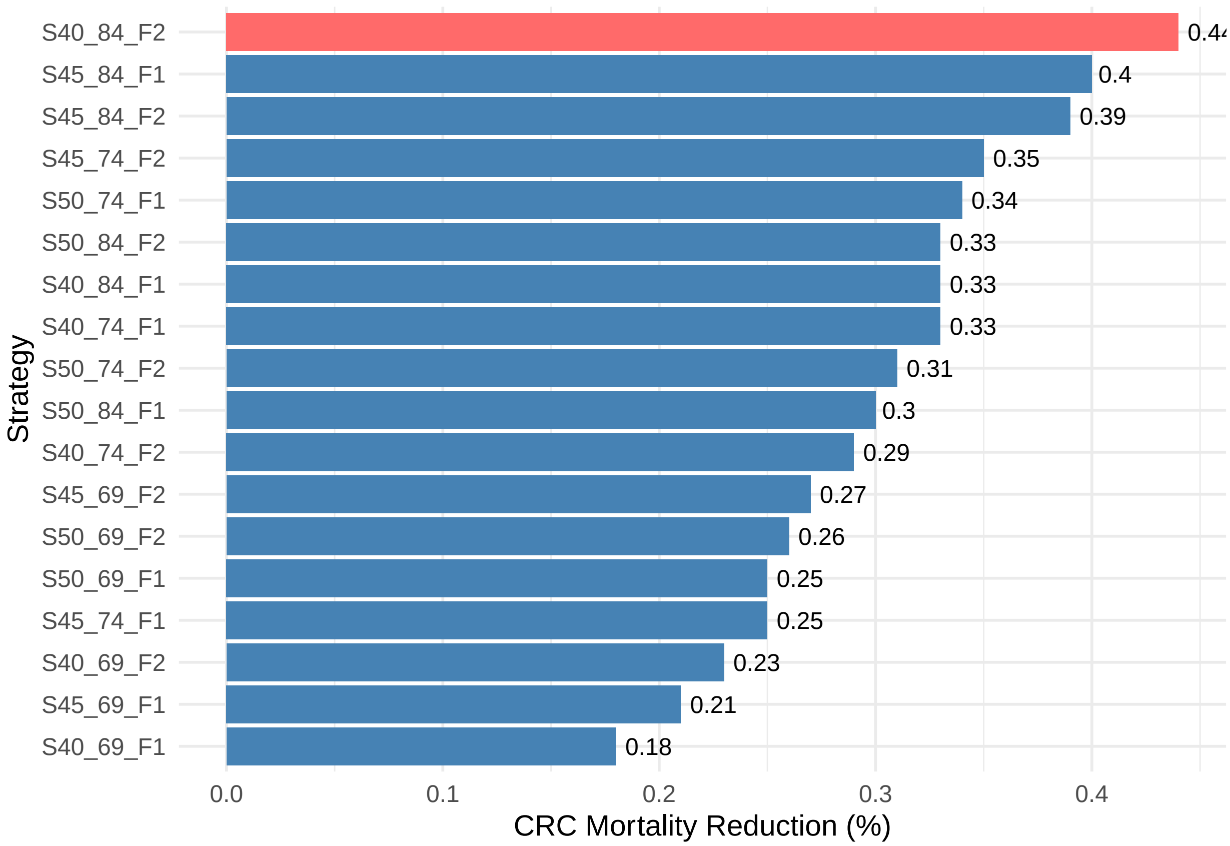



```r
ggsave("./pic/cost_by_strategy.png")
```

```
## Saving 6.5 x 4.5 in image
```

2.7 Mortality reduction by strategy

```r
ggplot(avg_df, aes(x = reorder(strategy, mortalityReduction_avg),
                   y = mortalityReduction_avg))+
  geom_col(aes(fill = mortalityReduction_avg == max(mortalityReduction_avg)))+
  scale_fill_manual(values = c("steelblue", "#FF6A6A"),
                    guide = "none")+
  coord_flip()+
  geom_text(aes(label = round(mortalityReduction_avg, 3)), hjust = -0.2, size = 3)+
  labs(
    x = "Strategy",
    y = "CRC Mortality Reduction (%)"
  )+
  theme_minimal()
```

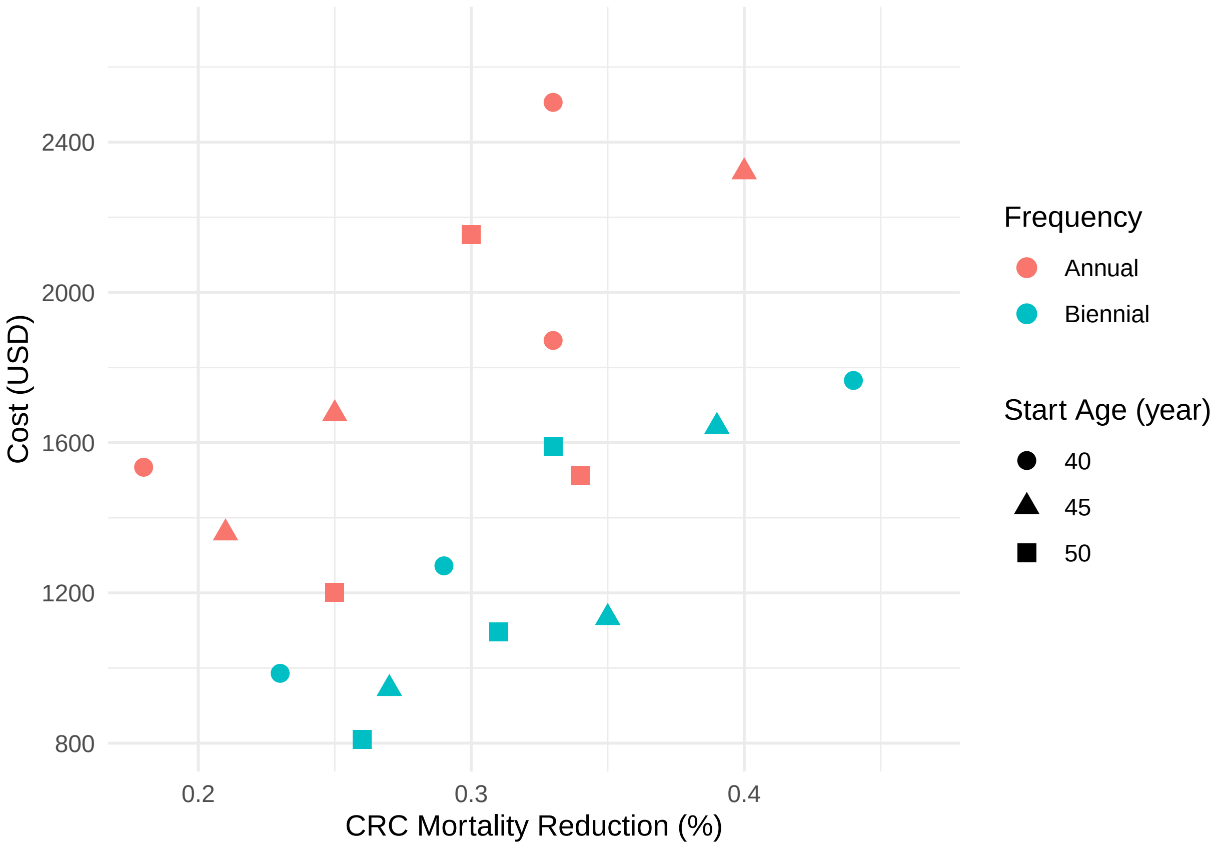



```r
ggsave("./pic/MRreduction_by_strategy.png")
```

```
## Saving 6.5 x 4.5 in image
```

2.8 Tradeoff: Cost vs Mortality (+ Frequency & Start Age)

```r
ggplot(avg_df, aes(x = mortalityReduction_avg, y = cost_avg,
                   colour = factor(frequency),
                   shape = factor(start_age)))+

  geom_point(size = 3)+

  scale_x_continuous(expand = expansion(mult = c(0.05, 0.15)))+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.15)))+

  coord_cartesian(clip = "off")+

  labs(
    x = "CRC Mortality Reduction (%)",
    y = "Cost (USD)",
    colour = "Frequency",
    shape = "Start Age (year)"
  )+
  theme_minimal()
```

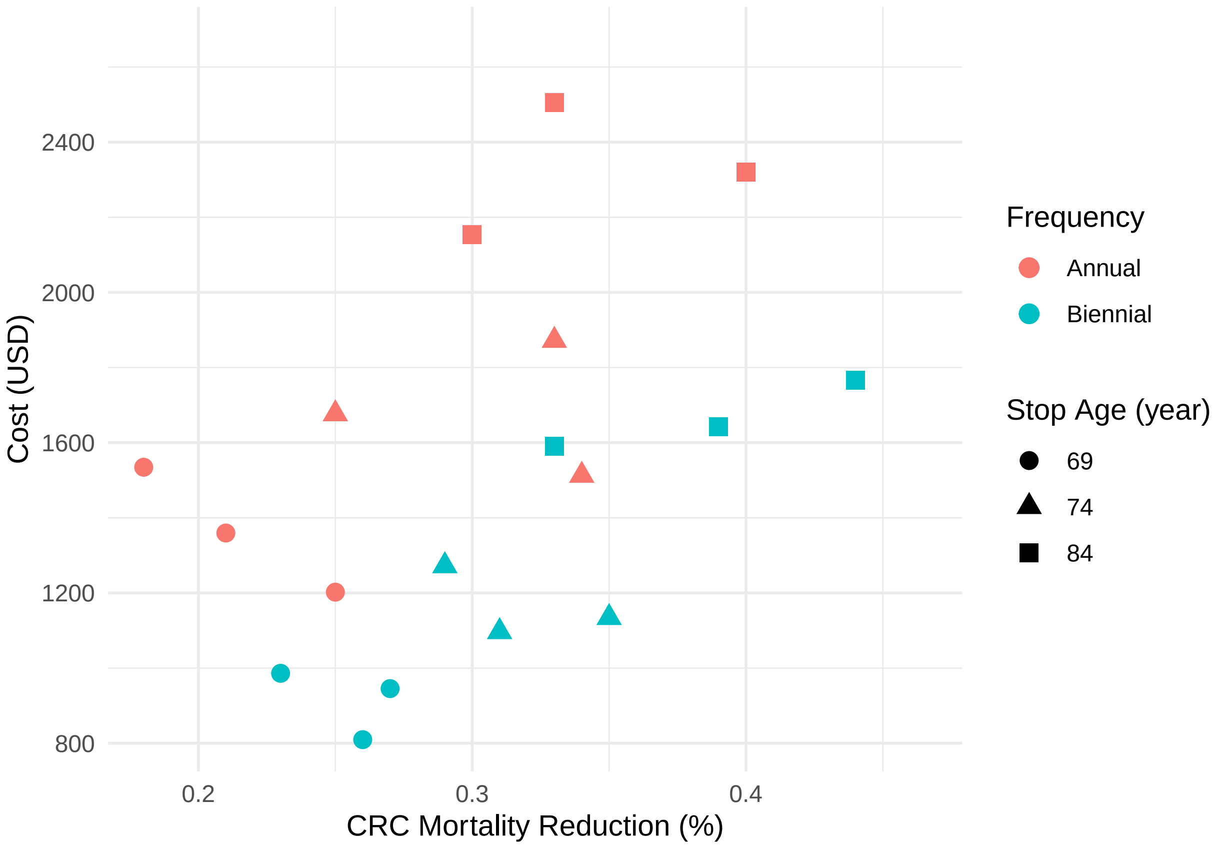



```r
ggsave("./pic/cost_MR_freq_startAge.png")
```

```
## Saving 6.5 x 4.5 in image
```

2.9 Tradeoff: Cost vs Mortality (+ Frequency & Stop Age)

```r
ggplot(avg_df, aes(x = mortalityReduction_avg, y = cost_avg,
                   colour = factor(frequency),
                   shape = factor(stop_age)))+

  geom_point(size = 3)+

  scale_x_continuous(expand = expansion(mult = c(0.05, 0.15)))+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.15)))+

  coord_cartesian(clip = "off")+

  labs(
    x = "CRC Mortality Reduction (%)",
    y = "Cost (USD)",
    colour = "Frequency",
    shape = "Stop Age (year)"
  )+
  theme_minimal()
```

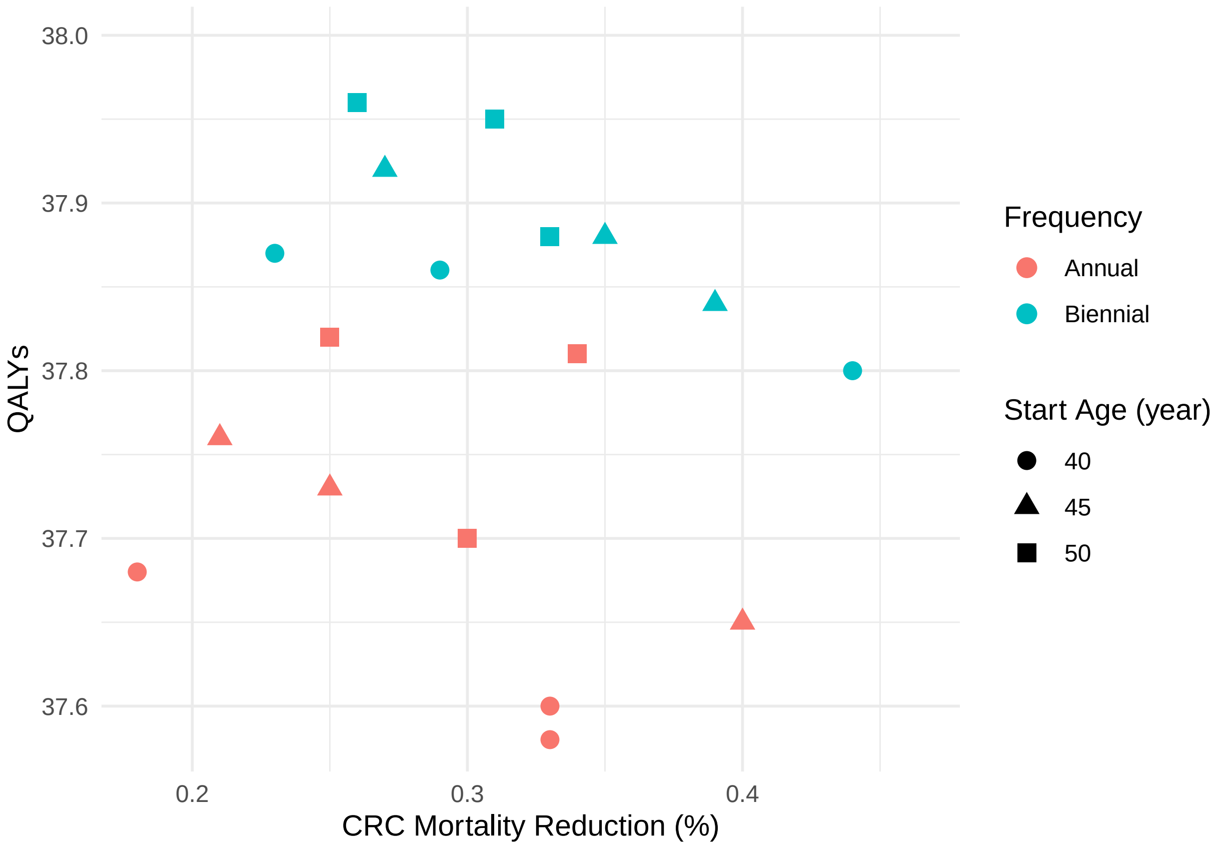



```
ggsave("./pic/cost_MR_freq_stopAge.png")
```

```
## Saving 6.5 x 4.5 in image
```

2.10 Tradeoff: QALYs vs Mortality (+ Frequency & Start Age)

```
ggplot(avg_df, aes(x = mortalityReduction_avg, y = qaly_avg,
                   colour = factor(frequency),
                   shape = factor(start_age)))+

  geom_point(size = 3)+

  scale_x_continuous(expand = expansion(mult = c(0.05, 0.15)))+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.15)))+

  coord_cartesian(clip = "off")+

  labs(
    x = "CRC Mortality Reduction (%)",
    y = "QALYs",
    colour = "Frequency",
    shape = "Start Age (year)"
  )+
  theme_minimal()
```

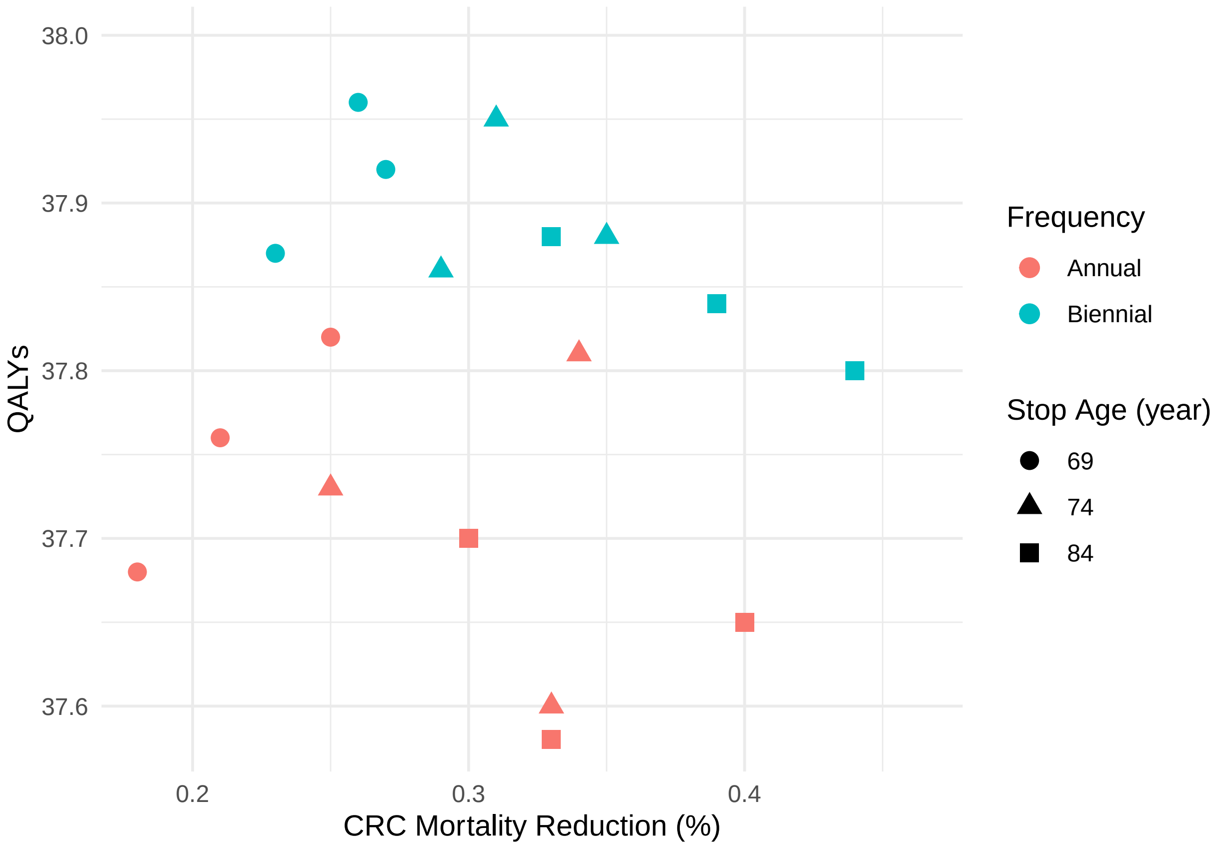



```
ggsave("./pic/qalys_MR_freq_startAge.png")
```

```
## Saving 6.5 x 4.5 in image
```

2.11 Tradeoff: QALYs vs Mortality (+ Frequency & Stop Age)

```
ggplot(avg_df, aes(x = mortalityReduction_avg, y = qaly_avg,
                   colour = factor(frequency),
                   shape = factor(stop_age)))+

  geom_point(size = 3)+

  scale_x_continuous(expand = expansion(mult = c(0.05, 0.15)))+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.15)))+

  coord_cartesian(clip = "off")+

  labs(
    x = "CRC Mortality Reduction (%)",
    y = "QALYs",
    colour = "Frequency",
    shape = "Stop Age (year)"
  )+
  theme_minimal()
```




```r
ggsave("./pic/qalys_MR_freq_stopAge.png")
```

```
## Saving 6.5 x 4.5 in image
```

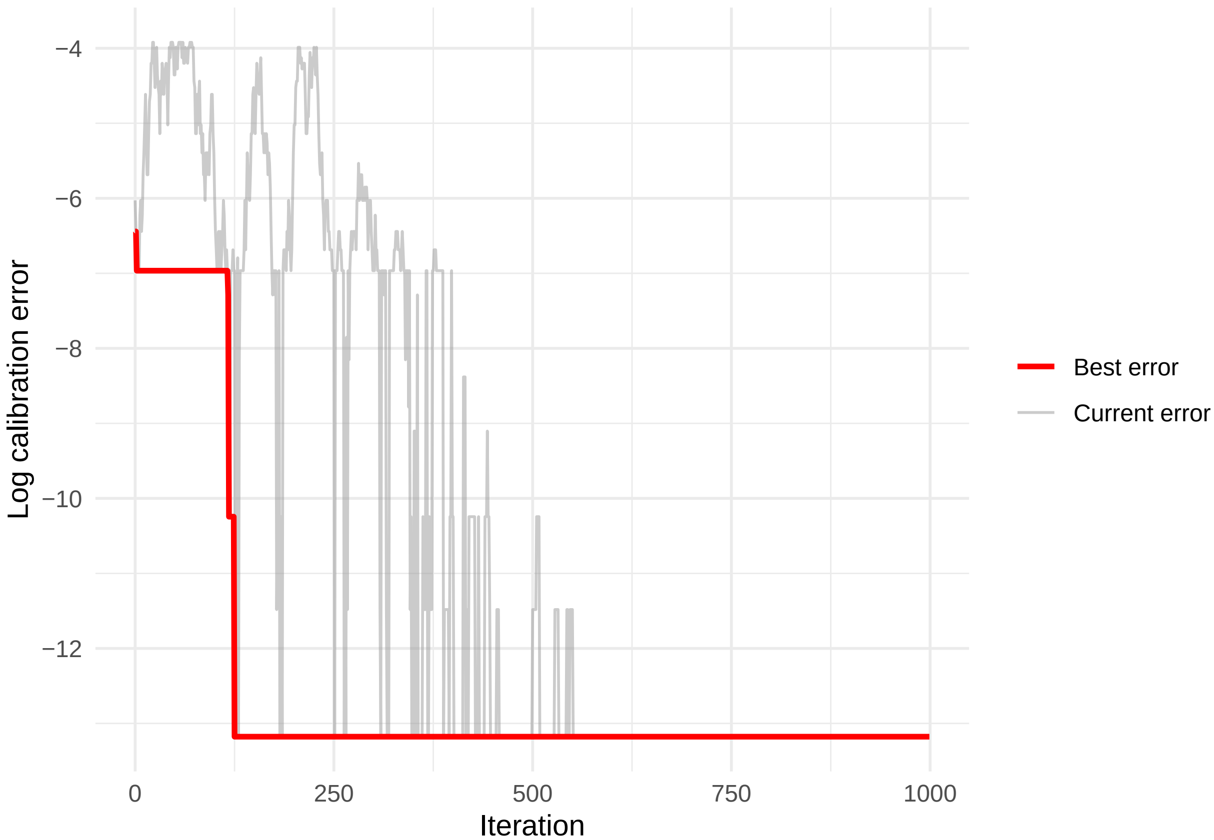
3. Calibration Plot

```r
calibration_trace <- read.csv("./FinalProject_Code/output/calibration_trace.csv")
head(calibration_trace, 3)
```

```
##   iteration currentEffect currentError bestEffect    bestError temperature
## 1         0     0.7236074  0.002409099  0.7236074 0.0015926690      1.0000
## 2         1     0.7312580  0.001592669  0.7236074 0.0015926690      0.9800
## 3         2     0.7773055  0.001592669  0.7773055 0.0009445754      0.9604
```

```r
ggplot(calibration_trace, aes(x = iteration)) +

  geom_line(
    aes(
      y = log(currentError + 1e-10),
      colour = "Current error"
    ),
    alpha = 0.5
  ) +

  geom_line(
    aes(
      y = log(bestError + 1e-10),
      colour = "Best error"
```

```
    ),
    linewidth = 1
  ) +

  scale_colour_manual(
    values = c(
      "Current error" = "grey60",
      "Best error" = "red"
    )
  ) +

  labs(
    x = "Iteration",
    y = "Log calibration error",
    colour = NULL
  ) +

  theme_minimal()
```




```
ggsave("./pic/calibration_trace.png")
```

```
## Saving 6.5 x 4.5 in image
```

4. Supplementary Table

4.1 Model input

```
input_data <- read.csv("./data/model_input.csv")
head(input_data, 3)
```

```
##   Age_group_years CRC_Incidence_Rate CRC_Mortality_Rate
## 1           40-44              35.79               9.15
## 2           45-49              60.32              15.44
## 3           50-54             108.23              26.85
##   All_Cause_Mortality_Rate
## 1                   182.65
## 2                   285.30
## 3                   431.36
```

```
input_tbl <- input_data|>
  gt()|>
  cols_label(
    Age_group_years = "Age Group (years)",
    CRC_Incidence_Rate = "CRC Incidence Rate (years)",
    CRC_Mortality_Rate = "CRC Mortality Rate (years)",
    All_Cause_Mortality_Rate = "All-Cause Mortality Rate (years)"
  )|>
  opt_table_font(font = "Times New Roman")|>
  tab_style(
    style = cell_text(
      weight = "bold",
      size = px(14),
      align = "center"
      ),
    locations = cells_column_labels()
  )|>
  tab_style(
    style = cell_text(
      size = px(14),
      align = "center"
    ),
    locations = cells_body()
  )

input_tbl
```

```
gtsave(data = input_tbl, filename = "input_tbl.png", path = "./table")
```

```
## file:////var/folders/10/f5shtmxx0jl9dg525f1vctqm0000gn/T//RtmpLwnkV1/file490615a375de.html screensho
```

4.2 Model calibrated parameters

```
calibrated_tbl <- data.frame(
  Parameter = c(
    "Target CRC mortality reduction",
    "Calibrated detection effect",
    "Model-predicted CRC mortality reduction",
    "Absolute calibration error",
```

Age Group (years)	CRC Incidence Rate (years)	CRC Mortality Rate (years)	All-Cause Mortality Ra
40-44	35.79	9.15	182.65
45-49	60.32	15.44	285.30
50-54	108.23	26.85	431.36
55-59	155.81	41.09	590.75
60-64	212.46	61.08	827.85
65-69	260.08	84.49	1184.23
70-74	306.94	118.33	1769.17
75-79	454.23	210.84	3315.64
80-84	491.46	312.24	5651.01

```
    "Optimisation method",
    "Number of iterations"
  ),

  Value = c(
    0.35,
    0.836,
    0.349,
    0.001,
    "Simulated annealing",
    1000
  )
)|>
  gt()|>
  opt_table_font(font = "Times New Roman")|>
  tab_style(
    style = cell_text(
      weight = "bold",
      size = px(14),
      align = "center"
      ),
    locations = cells_column_labels()
  )|>
  tab_style(
    style = cell_text(
      size = px(14)
    ),
    locations = cells_body()
  )|>
  cols_align(
    align = "center",
    columns = Value
  )

calibrated_tbl
```

```
gtsave(calibrated_tbl, filename = "calibrated_tbl.png", path = "./table")
```

```
## file:////var/folders/10/f5shtmxx0jl9dg525f1vctqm0000gn/T//RtmpLwnkV1/file49063ccbb67f.html screensho
```

Parameter	**Value**
Target CRC mortality reduction	0.35
Calibrated detection effect	0.836
Model-predicted CRC mortality reduction	0.349
Absolute calibration error	0.001
Optimisation method	Simulated annealing
Number of iterations	1000